# Seabed Mapping by Fusion of Multiple Sensors on an Autonomous Vehicle

Application to Bathymetry and Underwater Digital Terrain Models

English guide to the PhD thesis of Ridha Fezzani
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**About this document.** The original thesis is in French. Chapter 4, which describes the belief-function (Dempster–Shafer, "theory of evidence") fusion algorithm, is rendered here in full technical detail with every equation, then restated as a step-by-step procedure with the exact parameter values the author used. The other chapters are condensed into detailed English summaries. Equation numbers match the original so you can cross-reference the French PDF.

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## Abbreviations used

| French (original) | English |
|---|---|
| SMF – Sondeur MultiFaisceau | MBES – multibeam echosounder |
| SLI – Sondeur Latéral Interférométrique | ISSS – interferometric side-scan sonar |
| MNT – Modèle Numérique de Terrain | DTM – digital terrain model |
| sonde | sounding (one measured depth point) |
| fauchée | swath |
| portée | range |
| MCT – Modèle des Croyances Transférables | TBM – Transferable Belief Model |
| fonction de masse / structure de croyance | mass function / belief structure (BBA) |
| crédibilité / plausibilité | belief / plausibility |
| affaiblissement | discounting |
| probabilité pignistique | pignistic probability (BetP) |
| nonspécificité / désaccord | non-specificity / discord |
| facteur qualité | quality factor (QF) |
| RSB | SNR – signal-to-noise ratio |
| REA | Rapid Environment Assessment |

# 1 Abstract and general introduction (summary)

Over the last decade, autonomous underwater vehicles (AUVs) carrying many acoustic sensors have been used for Rapid Environment Assessment (REA), and they can operate covertly. For mapping wrecks, rocks and obstructions, side-scan sonar imagery is the usual tool. For bathymetry, the multibeam echosounder (MBES) is the most accurate instrument, but in shallow water its swath is narrow (about 2–4 times the altitude). Full coverage therefore takes many closely spaced lines, which an AUV's battery can rarely afford.

If the AUV also carries an *interferometric side-scan sonar* (ISSS), which covers a much wider swath but gives noisier soundings, the two sensors are both redundant and complementary. The thesis asks whether they can be fused into one more accurate and more complete DTM. To do this, the author develops a fusion algorithm based on **belief-function theory**. It accounts for the *uncertainty* (precision) of each sounding and the *reliability* of each sensor. The algorithm is first validated on simulated MBES/ISSS data, where the true seabed is known. It is then applied to real data from the DAURADE AUV (Reson 7125 MBES and Klein 5500 interferometric side-scan).

The manuscript has two parts. Part I (Chapters 1–3) covers underwater acoustics, the two sonar systems and how soundings plus their uncertainties are extracted, and a sonar signal simulator. Part II (Chapters 4–5) covers the fusion method and the results.

# 2 Part I summary – acoustics, sonars and simulation

### Chapter 1 – Underwater acoustics and bathymetric sonar systems

This chapter is background material. It covers the acoustic wave equation, intensity and power, the speed of sound in seawater (which depends on temperature, salinity and depth) and backscattering by the seabed. It also covers the *sonar equation* (echo level vs. noise and detection threshold), transmission losses (spherical spreading plus frequency-dependent absorption) and directivity gain. The depth-measuring systems are reviewed in turn: the single-beam echosounder and its footprint problem on irregular seabeds, multi-transducer "boom sweep" systems, the multibeam echosounder, the interferometric side-scan sonar and airborne bathymetric lidar. The chapter ends with the

error budget of a depth measurement. The total propagated uncertainty (TPU/TVU) combines sounder, motion-sensor, sound-speed, tide and positioning errors. The chapter also summarizes the IHO accuracy standards.

### Chapter 2 – The multibeam and interferometric side-scan sonars

**Multibeam (MBES).** A fan-shaped transmit beam is intersected with many narrow receive beams formed by *beamforming* on a linear array (with time-varying gain and array shading/weighting windows). In each beam the sounding is obtained from the arrival time and angle. Arrival time is estimated in one of two ways:

- *Amplitude detection* (best near nadir): threshold on the signal foot, maximum amplitude, or, most commonly, the **barycentre (centre of gravity)** of the echo envelope.
- *Phase detection* (best at oblique angles): the beam is split into two sub-arrays, and the zero crossing of their phase difference is found by **linear regression** of the phase ramp.

Angle of arrival can also be estimated by maximum-amplitude direction or high-resolution methods (MVDR, etc.).

**Uncertainty / quality factor.** For every sounding a quality factor is computed from the signal itself, following Lurton & Augustin (Ifremer):

$$QF = \log_{10}(qf) = \log_{10}\left(\frac{z}{\delta z}\right) = \log_{10}\left(\frac{t}{\delta t}\right) \tag{2.28}$$

For barycentre amplitude detection,

$$q_A = \frac{\sqrt{12}}{B\sqrt{\frac{4}{\pi}-1}}\,\frac{t_D\,N\,N_p}{T_{\text{eff}}\sqrt{S}} \tag{2.29}$$

where $t_D$ is the estimated arrival time, $N_p = \lfloor T f_s \rfloor$ (pulse length times sampling frequency), $T_{\text{eff}}$ the effective signal width (twice the square root of the second moment of the normalized envelope), $N = \lfloor f_s T_{\text{eff}} \rfloor$, $B \approx 1$ an envelope-shape factor, and $S$ a polynomial in $N, N_p$. For phase (zero-crossing) detection,

$$q_\phi = \frac{\alpha\,t_D}{\delta\Delta\phi}\,\frac{1}{\sqrt{\frac{(t_D-\bar{t})^2}{\sum_{i=1}^{N}(t_i-\bar{t})^2}+\frac{1}{N}}} \tag{2.31}$$

with $\delta\Delta\phi$ the phase standard deviation and $\alpha$ the regression slope. The chapter also covers survey planning (regular lines, cross-lines, infill lines) and MBES products: bathymetry, backscatter imagery and water column.

**Interferometric side-scan (ISSS).** The chapter covers side-scan geometry and resolution, two-antenna interferometry, and the sources of phase noise (baseline and footprint-shift decorrelation, plus SNR). Phase noise is reduced by *adaptive filtering* of the real and imaginary parts over a sliding, dynamic window. The $2\pi$ phase ambiguity is removed with the **Vernier method**, which uses three receivers and therefore three baselines with different spacings. The integer ambiguities $(k_1, k_2, k_3)$ are chosen per sample to minimize the summed Vernier mismatch between the pairs (eqs. 2.47–2.49). The ISSS quality factor (Lurton et al.) is

$$qf = \frac{z}{\delta z} = \frac{1}{\tan\theta\,\delta\gamma}, \qquad \delta\gamma = \sqrt{\delta\gamma_\psi^2 + \delta\gamma_T^2} \tag{2.51–2.52}$$

where $\delta\gamma_\psi = \frac{\lambda}{2\pi d\cos\gamma}\,\delta\Delta\psi$ comes from phase noise and $\delta\gamma_T$ from the pulse length. Side-scan survey planning must account for the blind zone under the vehicle, which requires interleaved lines.

**Why this matters for the fusion:** these quality factors give each sounding its own standard deviation $\sigma_i = \delta z$. That value becomes the width of the fuzzy number used in Chapter 4.

## Chapter 3 – Simulating MBES and ISSS signals

Raw signals are rarely recorded, and real data has no ground truth. The author therefore built a simulator based on the **bright-point (facet) model**. The seabed is meshed with triangular facets of about one wavelength. Each facet is a scatterer with a position, area, normal and seabed type. Its amplitude follows the APL-94 backscatter model for its seabed type (sand, mud, rock) and incidence angle. Facets hidden from the sonar are masked. The received baseband signal of each transducer is the sum of the delayed, attenuated contributions of all visible facets. The simulator produces beamformed MBES pings and ISSS interferograms. It was used to check that the quality factors of Chapter 2 match the true errors, for both the MBES and the ISSS.

# 3 Chapter 4 – The belief-function fusion algorithm (full detail)

## 4.1 Goals and the nature of the data

The fusion has to:

- build a DTM from the onboard bathymetric sensors, with 3-D points as both input and output;
- take the *imprecision* and *uncertainty* of the data into account;
- allow non-bathymetric information to be added later, such as information from acoustic imagery;
- output a *quality map* of the fused data, so that the robot can re-plan during the mission (adaptive survey).

This makes the task an *estimation problem with imprecise and uncertain data.*

The data from the two sounders have these imperfections:

1. **Imprecision.** Noise and processing errors affect every sounding. Imprecision is larger for the ISSS than for the MBES, differs from one sounding to the next, and grows with range.
2. **Uncertainty (reliability).** A sounding's final position also depends on other sensors: vehicle motion, immersion, sound-speed profile and tide. If any of these sensors fails, the sounding becomes unreliable.
3. **Coverage.** The MBES swath is at most about 4 times the altitude and very narrow on an AUV. The ISSS swath is much wider but has a blind zone at nadir.
4. **Resolution.** The ISSS has much finer across-track resolution (thousands of soundings per ping against a few hundred for the MBES).

## 4.2 Existing DTM fusion methods (state of the art)

Most DTM fusion work concerns land topography (InSAR, lidar, stereo-photogrammetry):

1. **Weighted average of elevations.** With $z = w_1 z_1 + w_2 z_2$, the weights come from the error variances (eq. 4.1). The result depends strongly on the choice of weights.
2. **Sparse representations.** A local patch is represented as a sparse combination of dictionary "terrain shapes", which rejects implausible shapes caused by outliers.
3. **Frequency-domain filtering.** The low frequencies of one DTM (coarse shape) are merged with the high frequencies of another (fine detail).
4. **Clustering.** Outliers are thresholded out using elevation and slope statistics. The remaining elevations in each cell are grouped with k-means, and the cluster mean is kept.

The only previous bathymetric fusion found by the author merged ISSS soundings with shape-from-shading depths from side-scan imagery, using fuzzy expert rules. Like the methods above, it ignores the uncertainty and reliability of each sounding and produces no quality measure.

Frameworks considered for uncertain information are Bayesian probability, fuzzy sets, possibility theory and Dempster–Shafer theory. Bayesian probability models only uncertainty and handles ignorance poorly. Dempster–Shafer theory models both *uncertainty* and *imprecision*, so it was chosen, in Smets' *Transferable Belief Model* (TBM) interpretation.

## 4.3 Belief-function theory: the building blocks

In the TBM, reasoning happens on two levels. At the *credal level*, information is modelled and revised. At the *pignistic level*, beliefs are turned into probabilities so that a decision can be made.

#### 4.3.1.1 Mass function (belief structure)

Let $\Omega$ be a finite *frame of discernment* and $2^\Omega$ the set of all its subsets. A *basic belief assignment* (BBA, or mass function) is a map $m : 2^\Omega \to [0, 1]$. It assigns to each subset $A$ the belief committed *exactly* to $A$, and to nothing more specific:

$$\sum_{A \subseteq \Omega} m(A) = 1 \tag{4.2}$$

- **Focal element:** any $A$ with $m(A) > 0$. The union of all focal elements is the *core*.
- **Categorical mass function:** a single focal element, $m(A) = 1$. If $A = \Omega$, this is the *vacuous* mass function, meaning total ignorance.
- **Bayesian mass function:** all focal elements are singletons ($m(A) > 0 \Rightarrow |A| = 1$), so $m$ is an ordinary probability distribution.
- **Normal mass function:** $m(\emptyset) = 0$. The mass $m(\emptyset)$ measures conflict. The TBM allows $m(\emptyset) > 0$ (*open world*, where the conflict may come from a hypothesis missing from $\Omega$). To return to a closed world, normalize:

$$m^*(A) = \begin{cases} \dfrac{m(A)}{1 - m(\emptyset)} & A \neq \emptyset \\ 0 & A = \emptyset \end{cases} \tag{4.3}$$

#### 4.3.1.2 Belief and plausibility

Belief is the total mass that *supports* $A$:

$$bel(A) = \sum_{\emptyset \neq B \subseteq A} m(B) \tag{4.4}$$

Plausibility is the total mass that does *not contradict* $A$:

$$pl(A) = 1 - bel(\overline{A}) = \sum_{A \cap B \neq \emptyset} m(B) \tag{4.5–4.6}$$

#### 4.3.1.3 Commonality and implicability

$$q(A) = \sum_{A \subseteq B} m(B), \qquad b(A) = \sum_{B \subseteq A} m(B) \tag{4.7–4.8}$$

These are used less often, but they simplify some formulas and their implementation.

#### 4.3.1.4 Discounting (reliability)

A source's reliability $\alpha \in [0, 1]$ is applied by moving part of its belief onto $\Omega$ (ignorance):

$$m_\alpha(A) = \alpha\, m(A) \qquad \forall A \subset \Omega,\ A \neq \Omega \tag{4.9}$$

$$m_\alpha(\Omega) = 1 - \alpha + \alpha\, m(\Omega) \tag{4.10}$$

Reliability can reflect, for example, how representative the training data is, or how reliable the sensor is in the situation being assessed.

### 4.3.2 Conjunctive and disjunctive combination

Sources are combined recursively. The **conjunctive rule** (the unnormalized basis of Dempster's rule) is

$$m_{1\cap 2}(A) = \sum_{B\cap C=A} m_1(B)\, m_2(C) \tag{4.11}$$

and the **disjunctive rule** is

$$m_{1\cup 2}(A) = \sum_{B\cup C=A} m_1(B)\, m_2(C) \tag{4.12}$$

The conjunctive rule can put mass on $\emptyset$, which measures the conflict between the sources:

$$m_{1\cap 2}(\emptyset) = \sum_{B\cap C=\emptyset} m_1(B)\, m_2(C) \tag{4.13}$$

In a closed world this mass must be removed by normalizing with (4.3). Doing so gives *Dempster's rule*, the *orthogonal sum* $\oplus$.

### 4.3.3 Pignistic transformation (decision)

To decide, each focal element's mass is shared equally among its elements (Smets):

$$BetP(\omega) = \sum_{A\ni\omega} \frac{m^*(A)}{|A|} \tag{4.14}$$

### 4.3.4 Uncertainty measures

Following Klir, a belief function contains two kinds of uncertainty: *non-specificity* (imprecision, lack of knowledge) and *conflict* (disagreement).

- **Non-specificity** generalizes the Hartley measure $U(A) = \log_2 |A|$:

$$N(m) = \sum_{\emptyset\neq A\subseteq\Omega} m(A) \log_2 |A| \tag{4.15}$$

  It satisfies $0 \leq N(m) \leq \log_2 |\Omega|$. It is 0 when $m$ is a probability distribution and maximal at total ignorance $m(\Omega) = 1$. A larger value means the masses are less informative.

- **Discord**, which generalizes Shannon entropy:

$$D(m) = -\sum_{A\subseteq\Omega} m(A) \log_2 BetP(A) \tag{4.16}$$

- **Strife**, Klir's preferred conflict measure:

$$S(m) = -\sum_{A\subseteq\Omega} m(A) \log_2 \sum_{B\subseteq\Omega} m(B) \frac{|A\cap B|}{|A|} \tag{4.17}$$

  Here $|A\cap B|/|A|$ is the degree to which $A$ is a subset of $B$ ("subsethood" in fuzzy-set theory).

- **Total uncertainty** is a linear combination of non-specificity and conflict. In Chapter 5 it is $N + S$.

## 4.4 The bathymetric fusion model (EVEREG)

The model chosen is **EVEREG (EVidential REGression)** by Petit-Renaud & Denoeux (2004), originally built for non-parametric regression. It extends belief functions to **fuzzy focal elements**: a focal element is a fuzzy set (here a fuzzy number) instead of a crisp set. This combines the strength of fuzzy sets for imprecise data with the richer credal framework. Below, the model is adapted to bathymetry.

### 4.4.1 Building the mass functions

The input is the set $Z$ of all soundings from the MBES and the ISSS over the same area. The goal is a wider and more accurate global DTM, plus a quality factor for every fused sounding. First, a **regular grid** is defined over the survey area, and the depth is estimated at each grid node.

Each sounding $i$ of each sensor is described by:

- $y_i$: its 2-D horizontal position;
- $z_i$: its depth (vertical position);
- $\sigma_i$: the standard deviation of the acoustic measurement, estimated together with the sounding (from the quality factor of Chapter 2);
- $p_i \in [0, 1]$: the sensor's reliability when the sounding was taken. It can model the sounder's measurement conditions, or how well the other sensors that contribute to the sounding were working.

Each sounding $(y_i, z_i, \sigma_i, p_i)$ becomes a pair $(y_i, m_i)$ with the mass function

$$m_i(F_i) = p_i, \qquad m_i(Z) = 1 - p_i \tag{4.18}$$

Here $F_i$ is a fuzzy membership function (triangular, trapezoidal, Gaussian, ...). With a Gaussian fuzzy number,

$$F_i(u) = \exp\left(-\frac{1}{2}\frac{(u - z_i)^2}{\sigma_i^2}\right), \qquad u \in \mathbb{R} \tag{4.19}$$

Each $m_i$ therefore encodes the depth $z_i$, its precision $\sigma_i$ and its reliability $p_i$. The mass $1 - p_i$ on the whole domain $Z$ represents "this sounding may be meaningless".

### 4.4.2 Discounting by distance

Consider the evidence $I = \{e_i = (y_i, m_i),\ i = 1, \ldots, n\}$ and a grid node at horizontal position $y$. If $y_i$ is close to $y$ (in some metric $\Delta$), the unknown depth $z$ should be close to $z_i$, and $e_i$ should weigh heavily. If $y_i$ is far from $y$, $e_i$ says little about $z$. Each mass function is therefore discounted according to distance:

$$m_z[y, e_i](A) = \begin{cases} m_i(A)\,\phi\big(\Delta(y - y_i)\big) & A \in \mathcal{F}(m_i) \setminus \{Z\} \\ 1 - \phi\big(\Delta(y - y_i)\big) & A = Z \\ 0 & \text{otherwise} \end{cases} \tag{4.20}$$

where $\phi : \mathbb{R}^+ \to [0, 1]$ is decreasing with

$$\phi(0) \leq 1, \qquad \lim_{\Delta \to \infty} \phi(\Delta) = 0 \tag{4.21}$$

When $y \approx y_i$, the discount rate $1 - \phi$ tends to 0 and $m_z[y, e_i] \approx m_i$. When $y_i$ is far away, $m_z[y, e_i](Z) \approx 1$, the vacuous belief structure (total ignorance). *[Translator's note: The French text says the mass tends towards "l'ensemble vide (Z)", literally "the empty set (Z)". The intended meaning is the* vacuous *structure, with all mass on Z.]*

### 4.4.3 Combining the mass functions

All the discounted structures $m_z[\cdot\,|e_i]$ are then combined to obtain their joint effect on the estimate of $z$. The **conjunctive rule** is used because the vacuous structure is its neutral element: soundings far from the node ($m_z(Z) \approx 1$) then have no influence on the estimate. For each sensor,

$$m_z[y, I] = \bigcap_{i=1}^{k} m_z[y, e_i] \tag{4.22}$$

where only the $k$ **nearest neighbours** $\{y_i\}_{i=1}^{k}$ of the node are used. This makes combination much faster without changing the result, since distant soundings contribute almost nothing anyway.

Finally, the belief structures of the two sounders are combined with **Dempster's orthogonal rule** (normalized conjunctive combination):

$$m = m^1 \oplus m^2 \quad \text{(MBES } \oplus \text{ ISSS)}.$$

With fuzzy focal elements, intersection is the **fuzzy intersection** (pointwise minimum), $C(z) = \min(A(z), B(z))$. The degree of overlap is the *height* $\mathrm{Int}(A, B) = \sup_{z\in Z} \min(A(z), B(z))$.

### 4.4.4 Computing the output

Because the focal elements are fuzzy, the pignistic *density* lives on the interval bounded by the minimum and maximum depths of the survey area:

$$BetP[y, I](z) = \sum_{A\in\mathcal{F}(m_z^*[y,I])} m_z^*[y, I](A)\,\frac{A(z)}{|A|} \tag{4.23}$$

where the cardinality of a fuzzy set on the discretized domain $Z$ is $|A| = \sum_{z\in Z} A(z)$, and $m^*$ is the normalized structure.

**Normalization of fuzzy structures (Yager's SNP).** Conjunctive combination can produce sub-normal fuzzy focal elements, with height $< 1$. Yager's *smooth normalization procedure* extends Dempster normalization to this case. For an unnormalized $m$ with focal elements $A \in \mathcal{F}(m)$, the normalized focal elements $B$ and masses are

$$\begin{cases} B(z) = \dfrac{A(z)}{h(A)} & \forall z \in Z \\ m^*(B) = \dfrac{\sum_{A^*=B} m(A)\,h(A)}{\sum_{A\in\mathcal{F}(m)} m(A)\,h(A)} \end{cases} \tag{4.24}$$

Here $h(A)$ is the height of fuzzy number $A$ and $A^*$ is its rescaled version $B$. The masses are redistributed over all focal elements, and lower-height focal elements (weaker agreement) lose weight. If every focal element has the same height, $m^*(A^*) = m(A)$.

**Point estimate.** When the focal elements are crisp numbers $\{z_i\}$, the pignistic expectation reduces to

$$\hat{z} = \sum_{i=1}^{n} m_z^*[y, I](\{z_i\})\,z_i + m_z^*[y, I](Z)\,\bar{z}, \qquad \bar{z} = \frac{z_{\min} + z_{\max}}{2} \tag{4.25}$$

This is a mass-weighted average of the soundings. The ignorance mass goes to the mid-range depth. In the general fuzzy case, the estimate uses each focal element's **centre of gravity**:

$$z_A^* = \frac{\sum_u u\,A(u)}{\sum_u A(u)}, \qquad \hat{z}(y) = \sum_{A\in\mathcal{F}(m_z^*[y,I])} m_z^*[y, I](A)\,z_A^* \tag{4.26}$$

**Confidence interval.** Use the quantiles of the pignistic density. More generally, decompose each fuzzy focal element into $\alpha$-cuts $A^\alpha$. For every $\alpha \in [0,1]$, the lower and upper expectations are

$$\underline{\hat{z}}^\alpha(y) = \sum_A m_z[y,I](A) \inf_{z \in A^\alpha} z, \qquad \overline{\hat{z}}^\alpha(y) = \sum_A m_z[y,I](A) \sup_{z \in A^\alpha} z \tag{4.27–4.28}$$

In Chapter 5, the interval used is the gap between the **1st and 9th deciles** of $BetP$.

### 4.4.5 Tuning and speed-ups

**Distance.** Since positions are numeric, use a generalized Euclidean distance

$$\Delta = \|y - y_i\| = \left[(y-y_i)^T N (y-y_i)\right]^{1/2} \tag{4.29}$$

with $N$ symmetric positive-definite.

**Dissimilarity function.** Denoeux recommends

$$\phi(\Delta) = \alpha \exp(-\gamma \Delta^2), \qquad 0.9 < \alpha < 1,\ \gamma > 0 \tag{4.30}$$

Petit-Renaud uses

$$\phi(|y - y_i|) = 0.99 \exp\left(-\frac{(y-y_i)^2}{\theta^2}\right) \tag{4.31}$$

**Optimizing $\theta$ by leave-one-out cross-validation.** Each input sounding $z_i$ is predicted from the other $n-1$ soundings $I^{-i}$ with (4.20) and (4.22), which gives $\hat{m}(\cdot|y_i,\theta) = m_{z_i}[y_i, I^{-i}, \theta]$. Its error against the true $m_i$ is averaged over all soundings:

$$C(\theta) = \frac{1}{N} \sum_{i=1}^{n} C\big(m_i, \hat{m}(\cdot|y_i,\theta)\big), \qquad \hat{\theta} = \arg\min_\theta C(\theta) \tag{4.32–4.33}$$

For fuzzy focal elements, the error criterion generalizes the squared Hausdorff distance between normalized fuzzy sets:

$$C(m_i, \hat{m}(\cdot|y_i,\theta)) = \sum_{F \in \mathcal{F}(m_i^*)} \sum_{F' \in \mathcal{F}(\hat{m}_i^*(\cdot|y))} m_i^*(F)\, \hat{m}_i^*(F'|y)\, \tilde{d}(F,F') \tag{4.34}$$

$$\tilde{d}(F,F') = \int_0^1 h_2(F_\alpha, F'_\alpha)\, d\alpha \tag{4.35}$$

$$h_2(F_\alpha, F'_\alpha) = \max\{(F_{\alpha-} - F'_{\alpha-}),\ (F_{\alpha+} - F'_{\alpha+})\} \tag{4.36}$$

where $F_\alpha$ is the $\alpha$-cut of $F$, with bounds $F_{\alpha+} = \sup$ and $F_{\alpha-} = \inf$.

**Computational cost.** Conjunctive combination has exponential complexity. There are two remedies (Denoeux): use only the $k$ nearest neighbours of $y$ (the option adopted here), or build belief structures from a small set of prototypes. A third option is to **merge similar fuzzy focal elements** iteratively during combination. Two focal elements $A, B$ at step $t$ are merged when their proximity

$$I(A,B) = \frac{m^t(A) + m^t(B)}{m^t(A)\, m^t(B)} \frac{S(A,B)}{1 - S(A,B)}, \qquad S(A,B) = \frac{|A \cap B|}{|A \cup B|} \tag{4.37–4.38}$$

is the highest. They are replaced by their mass-weighted average

$$C(z) = \frac{m^t(A) A(z) + m^t(B) B(z)}{m^t(A) + m^t(B)} \tag{4.39}$$

The merging stops once the maximum of $I$ reaches a threshold.

**Worked example (from the thesis)**

The goal is to estimate the depth at $y = (11.5, 0)$ from three nearby soundings with Gaussian fuzzy numbers (mean, spread):

1. $A = \text{Gauss}(20, 0.1)$, $m_1(A) = 0.9$, $y_1 = (11, 0)$
2. $B = \text{Gauss}(19.4, 0.1)$, $m_2(B) = 0.9$, $y_2 = (12, 0)$
3. $C = \text{Gauss}(20.4, 0.4)$, $m_3(C) = 0.9$, $y_3 = (11, 0)$ (less precise)

The frame is $[18, 23]$ and the dissimilarity parameters are $\alpha = 0.99$, $\gamma = 1$ in (4.30). All three soundings are 0.5 from $y$, so they are discounted equally. After combination and normalization, the focal elements are the original fuzzy numbers and their pairwise intersections (Fig. 4.1). Since the masses are equal and the overlap between $A$ and $C$ is the largest, the pignistic density peaks in the $A \cap C$ region. The estimated depth (centre of gravity) is about **20.0** (Fig. 4.2), with a smaller secondary peak near 19.4 that comes from $B$.

## 4.5 Conclusion of Chapter 4

Belief functions are a good tool for incomplete information. The chosen model extends them to fuzzy sets, so it benefits from both theories. It also produces quality measures (non-specificity, strife, pignistic confidence intervals) for the fused result, which was a key goal.

# 4 The algorithm as a step-by-step procedure

This section restates Chapters 4–5 as one procedure, with the parameter values used in the thesis.

**Evidential MBES + ISSS bathymetric fusion**

**Input:** soundings from each sensor $s \in \{\text{MBES}, \text{ISSS}\}$: $(y_i, z_i, \sigma_i, p_i)$.
**Output:** for each grid node $y$: fused depth $\hat{z}(y)$, confidence interval, non-specificity $N$, strife $S$, total uncertainty $N + S$.

1. **Extract soundings and their precision.**
   - MBES: beamforming → barycentre amplitude detection and phase linear-regression detection → quality factor (2.29)/(2.31) for each → keep, per beam, the detection with the highest QF.
   - ISSS: filter the real/imag parts of the phase differences of the 3 antenna pairs (sliding adaptive window, eq. 2.45) → Vernier angle of arrival → quality factor (2.51). The phase standard deviation is estimated over a sliding 40-sample window.
   - $\sigma_i = \delta z_i = z_i / q f_i$. Georeference everything, apply motion/sound-speed/tide corrections, and remove gross outliers.
2. **Encode each sounding** as a fuzzy belief structure: $m_i(F_i) = p_i$, $m_i(Z) = 1 - p_i$, with $F_i$ = Gaussian fuzzy number centred on $z_i$ with spread $\sigma_i$ (4.18–4.19). The thesis uses $p_i = 1$ for both sensors.
3. **Define a regular grid** over the area (step 20 cm in the thesis, about the MBES resolution) and discretize the depth domain $Z = [z_{\min}, z_{\max}]$.
4. **For each grid node $y$ and each sensor:**
   (a) find the $k$ nearest soundings ($k = 5$);
   (b) discount each with $\phi(d) = 0.99 \exp(-d^2/\theta^2)$, $\theta = 1$ (4.20, 4.31), so that mass $\phi\, p_i$ goes on $F_i$ and the rest on $Z$;
   (c) combine the $k$ structures with the fuzzy conjunctive rule (min-intersection of focal elements, product of masses), eq. (4.22).
5. **Combine the two sensors** with Dempster's rule $m = m^{\text{MBES}} \oplus m^{\text{ISSS}}$.
6. **Normalize** with Yager's smooth normalization (4.24): rescale every focal element to height 1, and reweight its mass by $m(A)\, h(A)$, renormalized.
7. **Decide:** compute the pignistic density (4.23). The fused depth is the mass-weighted centre of gravity (4.26). The confidence interval is the 1st–9th decile range of $BetP$.
8. **Qualify:** compute non-specificity (4.15) and strife (4.17) for each node, and total uncertainty $= N + S$. Optionally, post-process with CHARM to build the final DTM.

**Optional tuning:** choose $\theta$ (and $k$) by leave-one-out cross-validation (4.32–4.36). In the thesis this was too slow, so a single global $\theta$ was picked from the RMSE curve on simulated data.

**Intuition.** Each sounding says "the depth is around $z_i$ $\pm\sigma_i$, and I am trustworthy to degree $p_i$". Distance from the grid node weakens that statement. Intersecting the statements keeps the depths that most neighbours agree on. Precise MBES soundings produce narrow focal elements that dominate, and noisy ISSS soundings produce wide ones that mainly fill gaps. Where no data exists, mass stays on $Z$, so the interval becomes wide and non-specificity becomes high, which

honestly flags ignorance. Where soundings disagree, strife becomes high.

# 5 Chapter 5 – Results (summary)

## Simulated data

**Scenario.** The simulated seabed has three zones: sand ripples, flat mud and rock. One survey line at $y = 0$, 10 m above the seabed, was simulated with 300 pings of both sonars (configured as in Chapter 3). Processing followed step 1 above. As expected, ISSS soundings are noisier and have lower quality factors. Sand ripples make the MBES QF vary. The ISSS shows shadows at the end of its range and a blind zone at nadir.

**Parameters.** Grid step 20 cm; dissimilarity function (4.31); $p_i = 1$.

- *$k$:* RMSE against ground truth (50-ping central zone, $\theta = 1$) falls as $k$ grows (Fig. 5.6). This is because the resolutions (about 25 cm MBES, about 3 cm ISSS) make neighbouring soundings near-duplicates. $k = 5$ was chosen for speed.
- *$\theta$:* leave-one-out was too expensive, so one global value was chosen. RMSE is lowest for $\theta$ between 1 and 4 (Fig. 5.7), and $\theta = 1$ was chosen.

**Findings.**

- The fused map fills the ISSS shadows and nadir gap and improves the outer swath (Fig. 5.8).
- Non-specificity is lowest (most precise) where the MBES covers, and it clearly flags missing data in the ISSS shadows (Fig. 5.9).
- Strife (conflict) is high over the sand ripples, because of terrain variability and low SNR in the ripple troughs.
- Total uncertainty $N + S$ follows the same trend as the 1st–9th decile pignistic interval (Fig. 5.10).
- Against ground truth, the true residuals match the confidence interval well in mid-swath. The interval slightly over-estimates at the far range, because imprecise ISSS soundings widen the pignistic density (Fig. 5.11). The true seabed always lies inside the interval (Fig. 5.12).
- **CHARM comparison** (SHOM's robust hierarchical cleaning/modelling algorithm, threshold $1\sigma$): on the raw soundings, CHARM flags outliers even in mid-swath because of ISSS noise. On the fused soundings, outliers remain only at the far ISSS range. The DTM residual against truth shrinks from $[-0.96, 1.12]$ m (raw) to $[-0.34, 0.43]$ m (fused) (Fig. 5.13).

## Real data – DAURADE AUV

**Vehicle.** DAURADE ("Démonstrateur d'AUV de REA Discret") was built by ECA for the French defence procurement agency (DGA) and is operated by SHOM and GESMA. It is 5 m long, 0.7 m in diameter and weighs 1010 kg. It operates at 8–300 m depth, reaches up to 8 knots and has 10 h endurance at 4 knots.

**Klein 5500.** The Klein 5500 operates at 455 kHz ($\lambda \approx 3.3$ mm) with 20 kHz bandwidth. Each side has a 1.2 m imaging array of 12 transducers sampled at 22.75 kHz, and five beams are formed per side. Resolution is 3.75 cm × 10 cm at 75 m range and 4 knots. Each side also has two bathymetric receivers at $4\lambda$ and $6.5\lambda$. The depression angle was measured at 21.5° (nominal 20°). A third, synthetic receiver was built from four imaging transducers (5, 6, 8, 9 to port, referenced to transducer 1; 19–22 to starboard, referenced to transducer 15). Each transducer was corrected for the sequential multiplexing phase delay $\delta\varphi = 2\pi\,\nu_c/(32 \times 22750) = 5\pi/4$ per channel (eqs. 5.1–5.3).

**Reson SeaBat 7125.** The Reson operates at 400 kHz with a 128° × 0.5° swath and 0.5°×1° resolution. It forms 512 beams (equi-angle or equidistant) and samples at 34.48 kHz.

**Navigation.** DVL, PHINS inertial unit and pressure sensor were used, then post-processed with Delph INS between GPS fixes. The position was still inaccurate. Image registration was not attempted because the Klein images were too noisy at range (with spurious "fictitious ripples"

caused by interference). Instead, the ISSS lines were rigidly co-registered using manually picked control points (rocks, cables).

**Data.** North of the Bay of Douarnenez (Brittany), at 21–26 m depth, the area contains sand ripples and rocks. Two east–west lines 130 m apart were surveyed simultaneously by both sonars at 15 m altitude, covering $270 \times 600$ m. The MBES was logged as beamformed data so that QF could be computed. The Klein range was set to 75 m (100 and 150 m were too noisy). Only the ISSS swaths overlap.

**Results** (same parameters as for the simulated data, 20 cm grid):

- The precise MBES bathymetry is *not* degraded by the ISSS, and the Klein noise is reduced thanks to the outlier-robust $k$-NN scheme (Fig. 5.20).
- Non-specificity clearly separates MBES-covered from ISSS-only areas (Figs. 5.21, 5.23). ISSS precision is better in the rocky south, because of higher SNR.
- Strife is high at transitions between seabed types and on rock, where soundings vary and their fuzzy sets overlap little (Fig. 5.22).
- The pignistic decile interval confirms the non-specificity trend (Fig. 5.24). Along a south–north profile it widens with ISSS range (Fig. 5.25).

# 6 Conclusions and perspectives (summary)

The aim was not survey-grade charting but a *complete* bathymetry obtained *quickly* from an AUV, together with a *confidence map* for reconnaissance missions. The MBES is precise but narrow, while the ISSS is wide but noisy, with a nadir gap. Fusing the two when they run synchronously exploits their complementarity.

Contributions:

1. a bright-point MBES/ISSS signal simulator that provides ground truth;
2. full processing chains for both sonars, from raw data to soundings with a physical per-sounding error (Ifremer quality factor);
3. an evidential fuzzy fusion (EVEREG) that integrates per-sounding precision (fuzzy numbers) and per-source reliability. Using $k$ nearest neighbours gives robustness to outliers and reduces computation;
4. fuzzy outputs that allow quality qualification through the pignistic probability, non-specificity and strife. These measures were shown to track the true error on simulated data, and fusing *before* CHARM was shown to help.

All processing, from raw data to fusion, was written by the author in MATLAB without commercial software.

**Perspectives.**

- Computation time is the main drawback (512 MBES and about 4500 ISSS soundings per ping, 20 cm grid). The ISSS data could be subsampled while preserving the topography.
- Results depend heavily on horizontal positioning, so navigation should be corrected by image registration or SLAM.
- Shape-from-shading bathymetry from high-resolution imagery could be added as a third source.
- The uncertainty maps could drive *adaptive* surveying according to the mission's required precision.

**Publications listed:** R. Fezzani, B. Zerr, M. Legris, A. Mansour, Y. Dupas, "AUV swath bathymetric sensor data fusion for Rapid Environment Assessment", UACE 2015 (Chania, Greece; Young Scientist Award) and IEEE OES Beacon newsletter, June 2015. The author also won 1st prize in the IEEE Student Poster Competition at OCEANS'15. An IEEE JOE article was in

preparation.

**Key references for the algorithm:** Petit-Renaud & Denoeux (2004), EVEREG ("PRD04"); Petit-Renaud & Denoeux (1999), fuzzy belief structures ("PR99"); Denoeux (1997, 2000), evidential $k$-NN; Smets & Kennes (1994), TBM; Klir (2005), uncertainty measures; Yager (1996), smooth normalization; Lurton & Augustin / Lurton et al. (2012), quality factors; Debese et al. (2012), CHARM.